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Department of Mathematics

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Math 208 Major Exam I Term 251

Original exam date: 30 September 2025. Practice version: questions and choices shuffled.

Net time allowed: 90 minutes

1. By using a suitable substitution, we can transform the differential equation $(2y \cos^2 x - 2\sqrt{y}) dx + x \cos^2 x dy = 0$ into the linear differential equation

(a) $v' + \frac{1}{x}v = \frac{1}{x} \sec^2 x$

(b) $v' - \frac{1}{x}v = \frac{1}{x} \sec^2 x$

(c) $v' - \frac{2}{x}v = \frac{1}{x} \sec^2 x$

(d) $v' + \frac{2}{x}v = \frac{1}{x} \sec^2 x$

(e) $v' + \frac{2}{x}v = \frac{1}{x} \cos^2 x$

2. Which one of the following subsets is not a subspace of $\mathbb{R}^3$?

(a) The set of all vectors (x_1, x_2, x_3) such that $x_3 = 0$

(b) The set of all vectors (x_1, x_2, x_3) such that $x_1 + x_2 = 0$

(c) The set of all vectors (x_1, x_2, x_3) such that $x_1 + x_2 + x_3 = 0$

(d) The set of all vectors (x_1, x_2, x_3) such that $x_3 = x_1 + x_2$

(e) The set of all vectors (x_1, x_2, x_3) such that $x_3 = 1$

3. If the vectors $\mathbf{u} = (5, -4, 3)$, $\mathbf{v} = (-2, 0, 3)$ and $\mathbf{w} = (a, -8, 1)$ are linearly dependent, then $3a =$

(a) 42

(b) -40

(c) -42

(d) 36

(e) 40

4. The general solution of the differential equation $(2xy + 3y^2) dx - (2xy + x^2) dy = 0$ is given by

(a) $y^2 + xy = cx^4$

(b) $y^2 + xy = cx^3$

(c) $y^2 - xy = cx^2$

(d) $y^3 + xy = cx^2$

(e) $y^2 + 2xy = cx^3$

5. If the function $k(x)$ with $k(0) = 0$ makes $\frac{dy}{dx} = \frac{y \cos x + 2xe^y + 3}{k(x) - x^2e^y + 2x}$ an exact differential equation, then $k(x) =$

- (a) $\sin x + 2x$
- (b) $-\sin x + 4x$
- (c) $-\sin x + 3x$
- (d) $\sin x - 2x$
- (e) $-\sin x - 2x$

6. A general solution of the exact differential equation $(y^2 + 3x^2 - 2xy^3) dx - (1 - 2xy + 3x^2y^2) dy = 0$ is

- (a) $y - xy^2 - x^2y^3 - x^3 = c$
- (b) $y + xy^2 + x^2y^3 - x^3 = c$
- (c) $y - xy^2 + x^2y^3 - x^3 = c$
- (d) $y - xy^2 + x^2y^3 + x^3 = c$
- (e) $y - 2xy^2 + x^2y^3 + x^3 = c$

7. The general solution of the differential equation $x^2y'' + 3xy' = 2$ is given by (Note: A and B are constants, and $x > 0$)

- (a) $y(x) = \ln x + \frac{A}{x^2} + B$
- (b) $y(x) = \ln x + \frac{A}{x^2} + Bx$
- (c) $y(x) = 2 \ln x + \frac{A}{x^3} + B$
- (d) $y(x) = \ln x + \frac{A}{x^3} + B$
- (e) $y(x) = 2 \ln x + \frac{A}{x^2} + B$

8. The solution of the initial-value problem $(1 + x \cos^2 y + x + \cos^2 y) dx - x \sin y dy = 0$, $y(1) = 0$ is

- (a) $\ln |x| + 4x + \tan^{-1}(\cos y) = 1 + \frac{\pi}{4}$
- (b) $\ln |x| + 2x + \tan^{-1}(\cos y) = 1 + \frac{\pi}{4}$
- (c) $\ln |x| + 5x + \tan^{-1}(\cos y) = 1 + \frac{\pi}{2}$
- (d) $\ln |x| + 3x + \tan^{-1}(\cos y) = 1 + \frac{\pi}{3}$
- (e) $\ln |x| + x + \tan^{-1}(\cos y) = 1 + \frac{\pi}{4}$

9. If the solution space of the system

$$\begin{aligned}x_1 - 4x_2 - 3x_3 - 7x_4 &= 0 \\2x_1 - x_2 + x_3 + 7x_4 &= 0 \\x_1 + 2x_2 + 3x_3 + 11x_4 &= 0\end{aligned}$$

is the set of all linear combinations of the form $s\mathbf{u} + t\mathbf{v}$ where s, t are real numbers, then

- (a) $\mathbf{u} = (-5, -3, 0, -1)$ and $\mathbf{v} = (-1, -1, 1, 0)$
- (b) $\mathbf{u} = (-5, -3, 0, 1)$ and $\mathbf{v} = (1, -1, 1, 0)$
- (c) $\mathbf{u} = (-5, -3, 0, -1)$ and $\mathbf{v} = (1, 1, -1, 0)$
- (d) $\mathbf{u} = (-5, -3, 0, 1)$ and $\mathbf{v} = (-1, -1, 1, 0)$
- (e) $\mathbf{u} = (5, -3, 0, 1)$ and $\mathbf{v} = (-1, -1, 1, 0)$

10. At 4:00 pm, a thermometer reading 20°C is put into a freezer where the temperature is -10°C . If the reading is 5°C at 4:02 pm, then the reading at 4:06 pm is

- (a) -8.25°C
- (b) 0°C
- (c) -6.25°C
- (d) -7.25°C
- (e) -4.25°C

11. By making a suitable substitution, the differential equation $\frac{dy}{dx} = 1 + e^{y-x+5}$ can be transformed into the separable differential equation

- (a) $e^v dv = dx$
- (b) $e^v dv = 2 dx$
- (c) $e^{-v} dv = x dx$
- (d) $e^{-v} dv = dx$
- (e) $e^{-v} dv = 2 dx$

12. The general solution of the linear differential equation $(x^2 + 1)\frac{dy}{dx} + 3x^3y = 6x e^{-\frac{3}{2}x^2}$ is

- (a) $y(x) = [-6 + c(x^2 + 1)^{3/2}] e^{-\frac{3}{2}x^2}$
- (b) $y(x) = [-2 + c(x^2 + 1)^{3/2}] e^{-\frac{3}{2}x^2}$
- (c) $y(x) = [-3 + c(x^2 + 1)^{3/2}] e^{-\frac{3}{2}x^2}$
- (d) $y(x) = [-4 + c(x^2 + 1)^{3/2}] e^{-\frac{3}{2}x^2}$
- (e) $y(x) = [1 + c(x^2 + 1)^{3/2}] e^{-\frac{3}{2}x^2}$

13. A particle is moving in a straight line with acceleration $a(t) = t^2 \ln t$ and initial velocity $v(1) = 0$. The velocity at any time $t > 1$ is given by

(a) $\frac{t^4}{4} \ln t - \frac{t^4}{9} + \frac{1}{9}$

(b) $\frac{t^4}{4} \ln t - \frac{t^4}{8} + \frac{1}{8}$

(c) $\frac{t^3}{3} \ln t - \frac{t^3}{9} + \frac{1}{9}$

(d) $\frac{t^2}{2} \ln t - \frac{t^2}{8} + \frac{1}{8}$

(e) $\frac{t^3}{3} \ln t - \frac{t^3}{8} + \frac{1}{8}$

14. Let $\mathbf{v}_1 = (5, 3, 4)$, $\mathbf{v}_2 = (3, 2, 5)$ and $\mathbf{w} = (1, 0, -7)$ be three vectors in $\mathbb{R}^3$. If $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2$, then $ab =$

(a) 6

(b) 10

(c) -8

(d) -6

(e) 0

15. Let $y = x^m$ be a solution of the differential equation $xy^{(4)} + 6y''' = 0$. Then the sum of all values of m is equal to

(a) -1

(b) 0

(c) 3

(d) 1

(e) 2

Answer Key

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Q	Ans	Brief explanation
1	(a)	Bernoulli $n = \frac{1}{2}$, $v = \sqrt{y}$: $v' + \frac{1}{x}v = \frac{\sec^2 x}{x}$.
2	(e)	$x_3 = 1$ does not contain 0 .
3	(e)	$\det = 160 - 12a = 0$, so $3a = 40$.
4	(b)	$v = y/x$: $xv' = \frac{v^2+v}{2v+1}$, so $v^2 + v = cx$.
5	(e)	$M_y = \cos x + 2xe^y$, $N_x = -k' + 2xe^y - 2$, so $k' = -\cos x - 2$.
6	(c)	$F = xy^2 + x^3 - x^2y^3 - y$; multiply $F = C$ by -1 .
7	(a)	$p = y'$: $(x^3p)' = 2x$, $p = \frac{1}{x} + Cx^{-3}$.
8	(e)	Separable: $\frac{1+x}{x}dx = \frac{\sin y}{1+\cos^2 y}dy$; $C = 1 + \tan^{-1} 1$.
9	(d)	$x_2 = -x_3 - 3x_4$, $x_1 = -x_3 - 5x_4$.
10	(c)	$T = -10 + 30e^{-kt}$, $e^{-2k} = \frac{1}{2}$, $T(6) = -10 + \frac{30}{8}$.
11	(d)	$v = y - x + 5$: $v' = e^v$.
12	(b)	$\rho = e^{\frac{3}{2}x^2}(x^2 + 1)^{-3/2}$, $(\rho y)' = 6x(x^2 + 1)^{-5/2}$, $\rho y = -2(x^2 + 1)^{-3/2} + c$.
13	(c)	By parts: $\frac{t^3}{3} \ln t - \frac{t^3}{9} + C$; $v(1) = 0$ gives $C = \frac{1}{9}$.
14	(d)	$a = 2$, $b = -3$ (check $4a + 5b = -7$), so $ab = -6$.
15	(b)	$m(m-1)(m-2)(m+3) = 0$; sum = 0.